

UNIVERSIDAD EL BOSQUE

FACULTAD ING. DE SISTEMAS

SISTEMAS OPERACIONALES

RETO 10 — Seguridad operacional (Fail2Ban)

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RESUMEN:
El ejercicio consiste en proteger un servidor frente a intentos de acceso no autorizado, como el adivinar contraseñas por fuerza bruta. Para ello, se utiliza una máquina virtual con Ubuntu y la herramienta Fail2Ban, que analiza los registros del sistema y, cuando detecta que una misma dirección IP acumula varios intentos fallidos, la bloquea automáticamente mediante el cortafuegos. Se configuraron dos jails: una para el servicio SSH y otra para Nginx, que vigila la ruta /admin/, protegida con usuario y contraseña. Posteriormente, un segundo nodo que actúa como atacante realiza intentos fallidos contra ambos servicios, con el fin de comprobar que Fail2Ban los detecta y bloquea su dirección IP en los dos casos. A continuación, se simula un incidente en el que la IP legítima queda bloqueada por error, y se procede a diagnosticar la causa y desbloquearla manualmente sin desactivar la protección. Después, esa misma dirección se incorpora a whitelist, para verificar que, aun fallando el inicio de sesión repetidas veces, nunca vuelve a ser bloqueada. Cada etapa se respalda con los registros del sistema antes y después del bloqueo, explicando el patrón que Fail2Ban identificó. Finalmente, se compara Fail2Ban con Wazuh para justificar porqué una herramienta más liviana resulta suficiente para este caso de uso.

Introducción:

Contexto:

La seguridad de los sistemas operativos es un aspecto fundamental para proteger los servidores y los servicios que ofrecen frente a posibles accesos no autorizados. Uno de los riesgos más comunes son los ataques de fuerza bruta, en los cuales se realizan múltiples intentos de autenticación con diferentes combinaciones de credenciales para intentar ingresar a un sistema.

En este proyecto se aborda esta problemática mediante la implementación de Fail2Ban en un entrono Linux, una herramienta que permite supervisar los registros del sistema, identificar patrones de intentos fallidos de autenticación y aplicar bloqueos temporales sobre las direcciones IP que representan este comportamiento.

La implementación contempla la protección del servicios SSH y un segundo servicio mediante Nginx con autenticación HTTP Basic. Además, se busca comprobar el funcionamiento de los mecanismos de detección y bloqueo mediante pruebas realizadas desde un segundo nodo, asi como evaluar la capacidad de respuesta ante un incidente en el que la dirección IP legítima sea bloqueada accidentalmente.

De esta manera, el proyecto permite analizar como una herramienta de seguridad puede contribuir a reducir los intentos de acceso no autorizado y fortalecer la protección de los servicios de un servidor Linux.

Objetivos generales:

Implementar un mecanismo de protección en un servidor de Linux mediante fail2Ban, que permita detectar y bloquear automáticamente direcciones ip que realicen intentos repetitivos de acceso no autorizado, fortaleciendo la seguridad de los servicios expuestos.

Objetivos específicos:

- configurar Fail2Ban para supervisar los intentos de autenticación del servidor de SSH.
- Implementar una segunda Jail para proteger un servidor adicional mediante Nginx.

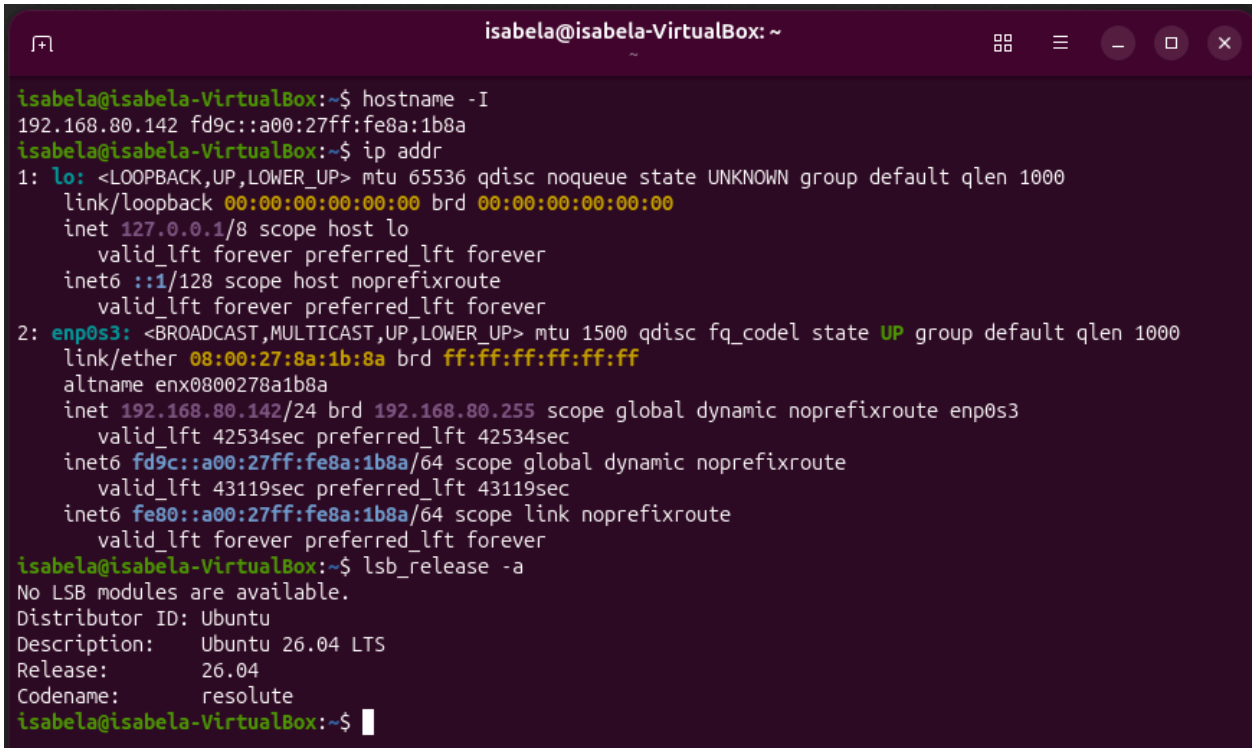
- Establecer una lista blanca (whitelist) de direcciones IP confiables.
- Configurar registros que permitan identificar y consultar los bloqueos realizados.
- Validar el funcionamiento de Fail2Ban mediante intentos de acceso desde un segundo nodo.
- Simular un incidente de bloqueo accidental y aplicar el procedimiento para recuperar el acceso sin desactivar la protección.

Alcance:
El proyecto comprende la implementación de un mecanismo de protección basado en Fail2Ban sobre un servidor ubuntu Linux. Se realizan la configuración de dos mecanismos de supervisión; uno dirigido al servidor SSH, encargado de permitir conexiones remotas, y otro asociado a Nginx, mediante autenticación HTTP Basic para restringir el acceso a un recurso web.

Asimismo. Se incluyen la configuracion de parámetros para controlar la cantidad de intentos fallidos, el periodo de detección y la duración de los bloqueos, junto con la definición de direcciones IP confiables y la generación de registros que permitan consultar los eventos de seguridad.

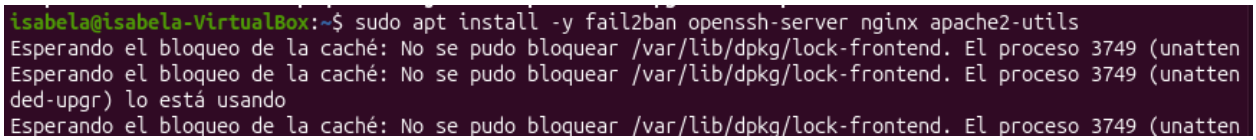
Para validar la implementación, se utiliza un segundo nodo que simula el comportamiento de un equipo atacante, desde el cual se realizan intentos fallidos de autenticación. También se contempla la simulación de un incidente de bloqueo accidental y la aplicación de un procedimiento para recuperar el acceso sin interrumpir el funcionamiento del mecanismo de protección.

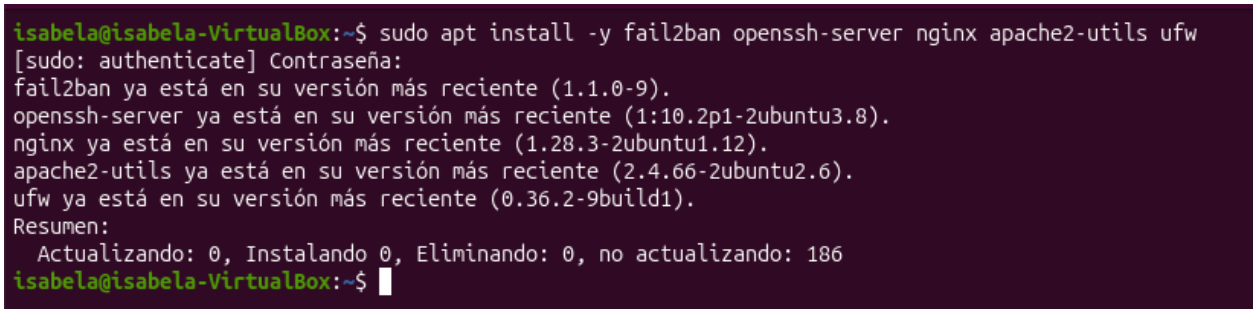
El proyecto se limita a un entorno académico de pruebas con máquinas virtuales y no contempla la implementación de una infraestructura empresarial distribuida, la integración de una plataforma centralizada de monitoreo como Wazuh ni la evaluación de servidores expuestos directamente a Internet.



INSTALACIONES

INSTALACION DE GNINX, FAIL2BAN Y UFW





VERSIONES

```
isabela@isabela-VirtualBox:~$ ssh -V
OpenSSH_10.2p1 Ubuntu-2ubuntu3.8, OpenSSL 3.5.5 27 Jan 2026
isabela@isabela-VirtualBox:~$ fail2ban-client --version
Fail2Ban v1.1.0
isabela@isabela-VirtualBox:~$ nginx -v
nginx version: nginx/1.28.3 (Ubuntu)
isabela@isabela-VirtualBox:~$ ufw version
ufw 0.36.2
Copyright 2008-2023 Canonical Ltd.
isabela@isabela-VirtualBox:~$
```

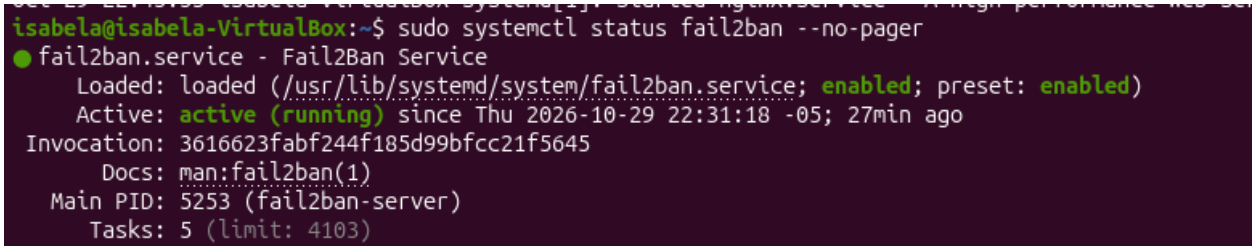
Instalación de las herramientas necesarias para implementar el sistema de protección.

Paso 4. Activar los servicios

```
isabela@isabela-VirtualBox:~$ sudo systemctl enable --now ssh nginx fail2ban
Synchronizing state of ssh.service with SysV service script with /usr/lib/systemd/systemd-sysv-install.
Executing: /usr/lib/systemd/systemd-sysv-install enable ssh
Synchronizing state of nginx.service with SysV service script with /usr/lib/systemd/systemd-sysv-install.
Executing: /usr/lib/systemd/systemd-sysv-install enable nginx
Synchronizing state of fail2ban.service with SysV service script with /usr/lib/systemd/systemd-sysv-install
.
Executing: /usr/lib/systemd/systemd-sysv-install enable fail2ban
Created symlink '/etc/systemd/system/sshd.service' → '/usr/lib/systemd/system/ssh.service'.
Created symlink '/etc/systemd/system/multi-user.target.wants/ssh.service' → '/usr/lib/systemd/system/ssh.se
rvice'.
Created symlink '/etc/systemd/system/multi-user.target.wants/nginx.service' → '/usr/lib/systemd/system/ngin
x.service'.
```

```
isabela@isabela-VirtualBox:~$ sudo systemctl status ssh
● ssh.service - OpenBSD Secure Shell server
     Loaded: loaded (/usr/lib/systemd/system/ssh.service; enabled; preset: enabled)
     Active: active (running) since Thu 2026-10-29 22:45:53 -05; 1min 45s ago
 Invocation: 7827e920d24d4a27b75837cd7b4fb004
TriggeredBy: ● ssh.socket
       Docs: man:sshd(8)
             man:sshd_config(5)
    Process: 7590 ExecStartPre=/usr/sbin/sshd -t (code=exited, status=0/SUCCESS)
   Main PID: 7595 (sshd)
```

```
isabela@isabela-VirtualBox:~$ sudo systemctl status nginx
● nginx.service - A high performance web server and a reverse proxy server
     Loaded: loaded (/usr/lib/systemd/system/nginx.service; enabled; preset: enabled)
     Active: active (running) since Thu 2026-10-29 22:45:53 -05; 2min 7s ago
 Invocation: ce9b72cd7c664ad0a83b10890f7b028c
       Docs: man:nginx(8)
    Process: 7589 ExecStartPre=/usr/sbin/nginx -t -q -g daemon on; master_process on; (code=exited, status>
    Process: 7593 ExecStart=/usr/sbin/nginx -g daemon on; master_process on; (code=exited, status=0/SUCCES>
   Main PID: 7597 (nginx)
```



Verificación del funcionamiento de los servicios SSH, Nginx y Fail2Ban.

Etapa 2. Configurar la red y SSH

Configurar el cortafuegos

```
isabela@isabela-VirtualBox:~$ sudo ufw allow OpenSSH
Regla añadida
Regla añadida (v6)
isabela@isabela-VirtualBox:~$ sudo ufw allow 80/tcp
Omitiendo adición de regla ya existente
Omitiendo adición de regla ya existente (v6)
isabela@isabela-VirtualBox:~$ sudo ufw status
Estado: activo

Hasta                      Acción      Desde
-----                      ------      -----
2224/tcp                   ALLOW       Anywhere
5405/udp                   ALLOW       Anywhere
80/tcp                     ALLOW       Anywhere
2049/tcp                   ALLOW       Anywhere
OpenSSH                    ALLOW       Anywhere
2224/tcp (v6)              ALLOW       Anywhere (v6)
5405/udp (v6)              ALLOW       Anywhere (v6)
80/tcp (v6)                ALLOW       Anywhere (v6)
2049/tcp (v6)              ALLOW       Anywhere (v6)
OpenSSH (v6)               ALLOW       Anywhere (v6)

isabela@isabela-VirtualBox:~$ sudo ufw enable
El cortafuegos está activo y habilitado en el arranque del sistema
isabela@isabela-VirtualBox:~$ sudo ufw status verbose
Estado: activo
Acceso: on (low)
Predeterminado: deny (entrantes), allow (salientes), disabled (enrutados)
Perfiles nuevos: skip

Hasta                      Acción      Desde
-----                      ------      -----
2224/tcp                   ALLOW IN    Anywhere
5405/udp                   ALLOW IN    Anywhere
80/tcp                     ALLOW IN    Anywhere
2049/tcp                   ALLOW IN    Anywhere
22/tcp (OpenSSH)           ALLOW IN    Anywhere
2224/tcp (v6)              ALLOW IN    Anywhere (v6)
```

Figura 4. Configuración del cortafuegos para permitir las conexiones SSH y HTTP necesarias.

Preparar SSH para las pruebas

Crear un usuario de pruebas

```
isabela@isabela-VirtualBox:~$ getent passwd usuario_prueba
usuario_prueba:x:1003:1003::/home/usuario_prueba:/bin/bash
isabela@isabela-VirtualBox:~$
```

El usuario anteriormente se habia creado, pero se evidencia de que existe

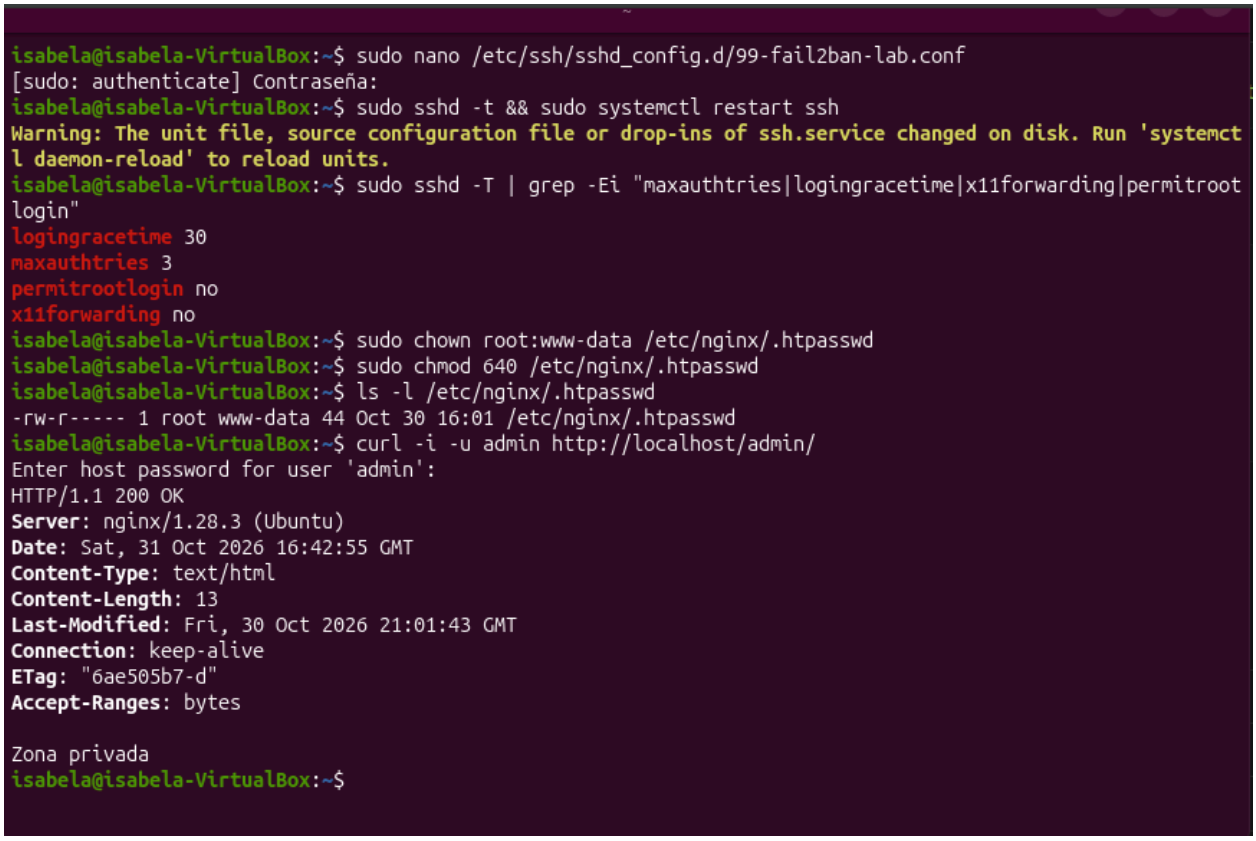
Verificación de usuarios y permisos

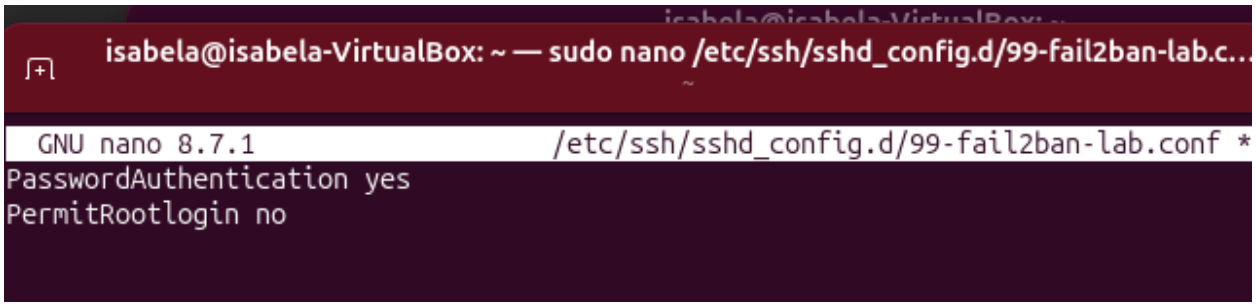
```
isabela@isabela-VirtualBox:~$ id usuario_prueba
uid=1003(usuario_prueba) gid=1003(usuario_prueba) groups=1003(usuario_prueba)
isabela@isabela-VirtualBox:~$  sudo -l -U usuario_prueba
[sudo: authenticate] Contraseña:
User usuario_prueba is not allowed to run sudo on isabela-VirtualBox.
isabela@isabela-VirtualBox:~$ ls -l /etc/nginx/.htpasswd
-rw-r----- 1 root www-data 44 Oct 30 16:01 /etc/nginx/.htpasswd
isabela@isabela-VirtualBox:~$ ssh root@localhost
root@localhost's password:
Permission denied, please try again.
root@localhost's password:
Permission denied, please try again.
root@localhost's password:
Received disconnect from 127.0.0.1 port 22:2: Too many authentication failures
Disconnected from 127.0.0.1 port 22
isabela@isabela-VirtualBox:~$ date; sudo sshd -T | grep -Ei "permitrootlogin|maxauthtries|logingracetime"
sáb 31 oct 2026 12:11:44 -05
logingracetime 30
maxauthtries 3
permitrootlogin no
isabela@isabela-VirtualBox:~$
```

Preparar SSH para las pruebas y hardening

```
isabela@isabela-VirtualBox:~$ id usuario_prueba
uid=1003(usuario_prueba) gid=1003(usuario_prueba) groups=1003(usuario_prueba)
isabela@isabela-VirtualBox:~$ sudo nano /etc/ssh/sshd_config.d/99-fail2ban-lab.conf
[sudo: authenticate] Contraseña:
isabela@isabela-VirtualBox:~$ sudo sshd -t
isabela@isabela-VirtualBox:~$ sudo systemctl restart ssh
isabela@isabela-VirtualBox:~$ sudo systemctl status ssh
● ssh.service - OpenBSD Secure Shell server
     Loaded: loaded (/usr/lib/systemd/system/ssh.service; enabled; preset: enabled)
     Active: active (running) since Thu 2026-10-29 23:26:28 -05; 16s ago
 Invocation: a7c9ed8039e546f2b81e97f9653a8737
TriggeredBy: ● ssh.socket
       Docs: man:sshd(8)
             man:sshd_config(5)
    Process: 8275 ExecStartPre=/usr/sbin/sshd -t (code=exited, status=0/SUCCESS)
   Main PID: 8278 (sshd)
      Tasks: 1 (limit: 4103)
     Memory: 1.2M (peak: 1.9M)
        CPU: 32ms
     CGroup: /system.slice/ssh.service
             └─8278 "sshd: /usr/sbin/sshd -D [listener] 0 of 10-100 startups"

oct 29 23:26:28 isabela-VirtualBox systemd[1]: Starting ssh.service - OpenBSD Secure Shell server...
oct 29 23:26:28 isabela-VirtualBox sshd[8278]: Server listening on 0.0.0.0 port 22.
oct 29 23:26:28 isabela-VirtualBox sshd[8278]: Server listening on :: port 22.
oct 29 23:26:28 isabela-VirtualBox systemd[1]: Started ssh.service - OpenBSD Secure Shell server.
isabela@isabela-VirtualBox:~$
```







Configuración de autenticación SSH para realizar las pruebas controladas de Fail2Ban.

Configurar Fail2Ban

Entender los archivos de configuración

```
isabela@isabela-VirtualBox:~$ ls /etc/fail2ban/
action.d        fail2ban.d      filter.d   jail.d      paths-arch.conf    paths-debian.conf
fail2ban.conf   fail2ban.local  jail.conf  jail.local  paths-common.conf  paths-opensuse.conf
isabela@isabela-VirtualBox:~$
```

Crear la jail de SSH

```
  GNU nano 8.7.1                                  /etc/fail2ban/jail.local
[DEFAULT]
bantime = 1h
findtime = 10m
maxretry = 5
ignoreip = 127.0.0.1/8 ::1

[sshd]
enabled = true
backend = systemd
port = ssh
maxretry = 5
findtime = 10m
bantime = 1h
```

```
isabela@isabela-VirtualBox:~$ sudo nano /etc/fail2ban/jail.local
isabela@isabela-VirtualBox:~$ sudo systemctl restart fail2ban
isabela@isabela-VirtualBox:~$ sudo fail2ban-client status
Status
|- Number of jail:      1
`- Jail list:   sshd
isabela@isabela-VirtualBox:~$ sudo fail2ban-client status sshd
Status for the jail: sshd
|- Filter
|  |- Currently failed: 0
|  |- Total failed:     0
|  `- Journal matches:  _SYSTEMD_UNIT=ssh.service + _COMM=sshd
`- Actions
   |- Currently banned: 0
   |- Total banned:     0
   `- Banned IP list:
isabela@isabela-VirtualBox:~$
```

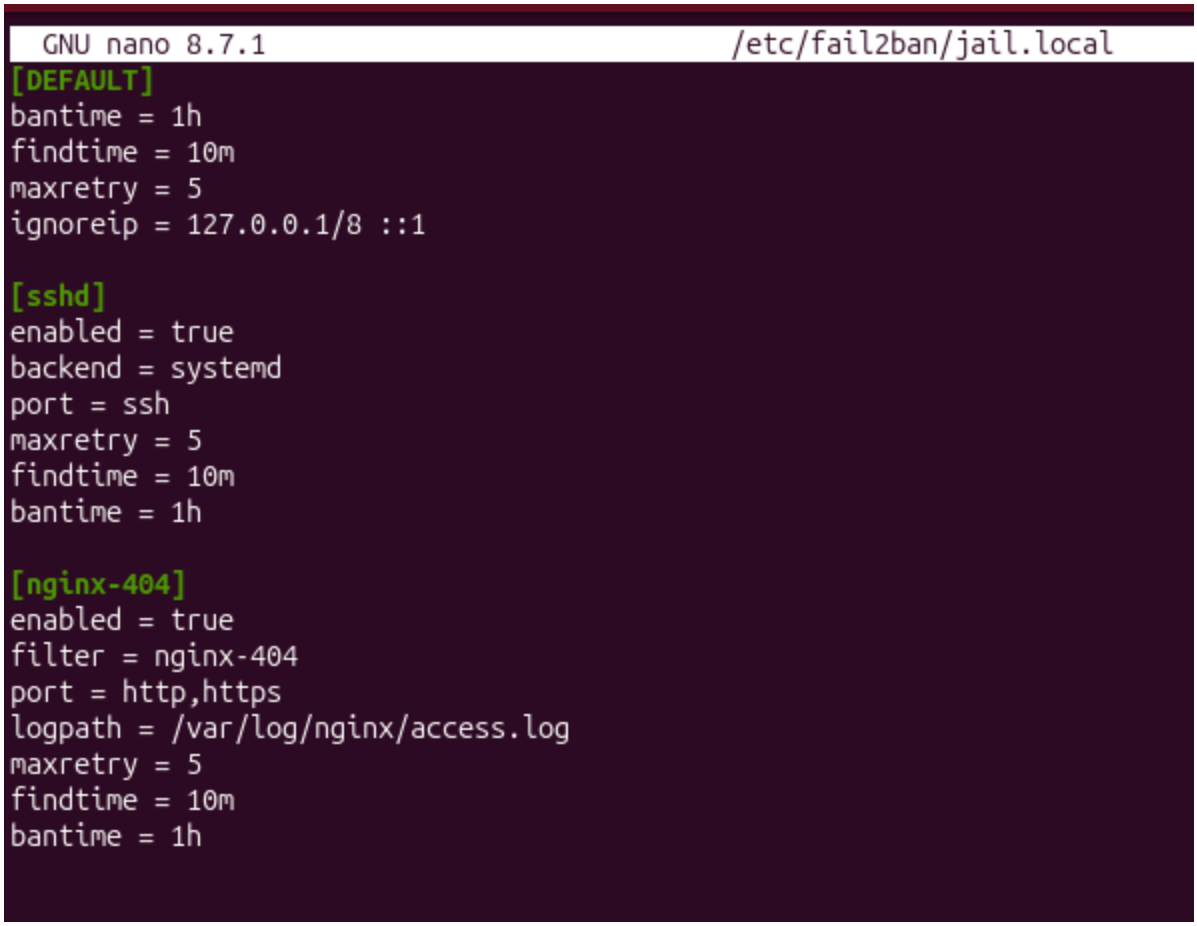
Configuración y activación de la jail SSH con los parámetros de detección y bloqueo definidos.

Configurar una segunda jail

Crear un filtro personalizado

```
 GNU nano 8.7.1                    /etc/fail2ban/filter.d/nginx-404.conf
[Definition]
failregex = ^<HOST> - .* "(GET|POST|HEAD) /admin[^"]*" (401|403|404) .*
ignoreregex =
```

Agregar la segunda jail


GNU nano 8.7.1 /etc/fail2ban/jail.local
[DEFAULT]
bantime = 1h
findtime = 10m
maxretry = 5
ignoreip = 127.0.0.1/8 ::1
[sshd]
enabled = true
backend = systemd
port = ssh
maxretry = 5
findtime = 10m
bantime = 1h
[nginx-404]
enabled = true
filter = nginx-404
port = http,https
logpath = /var/log/nginx/access.log
maxretry = 5
findtime = 10m
bantime = 1h

```
isabela@isabela-VirtualBox:~$ sudo nano /etc/fail2ban/jail.local
isabela@isabela-VirtualBox:~$ sudo systemctl restart fail2ban
isabela@isabela-VirtualBox:~$ sudo fail2ban-client status
Status
|- Number of jail:      2
`- Jail list:   nginx-404, sshd
isabela@isabela-VirtualBox:~$ sudo fail2ban-client status nginx-404
Status for the jail: nginx-404
|- Filter
|  |- Currently failed: 0
|  |- Total failed:     0
|  `- File list:        /var/log/nginx/access.log
`- Actions
   |- Currently banned: 0
   |- Total banned:     0
   `- Banned IP list:
isabela@isabela-VirtualBox:~$
```

```
isabela@isabela-VirtualBox:~$ sudo fail2ban-regex /var/log/nginx/access.log nginx-404

Running tests
=============

Use            jail : nginx-404
Use        log file : /var/log/nginx/access.log
Use        encoding : UTF-8


Results
=======

Failregex: 1 total
|-  #) [# of hits] regular expression
|   1) [1] ^<HOST> - .* "(GET|POST|HEAD) /admin[^"]*" (401|403|404) .*
`-

Ignoreregex: 0 total

Date template hits:
|- [# of hits] date format
|  [2] Day(?P<_sep>[-/])MON(?P=_sep)ExYear[ :]?24hour:Minute:Second(?:\.Microseconds)?(?: Zone offset)?
`-

Lines: 2 lines, 0 ignored, 1 matched, 1 missed
[processed in 0.04 sec]

|- Missed line(s):
|  ::1 - admin [30/Oct/2026:16:05:34 -0500] "GET /admin/ HTTP/1.1" 200 13 "-" "curl/8.18.0"
`-
isabela@isabela-VirtualBox:~$
```

Implementación de una segunda jail para detectar solicitudes sospechosas dirigidas al servidor web Nginx.

Configurar el registro de bloqueos

```
  GNU nano 8.7.1                                /etc/fail2ban/fail2ban.local *
[Definition]
logtarget = /var/log/fail2ban.log
loglevel = INFO
```

```
isabela@isabela-VirtualBox:~$ sudo tail -n 30 /var/log/fail2ban.log
2026-10-30 17:26:20,568 fail2ban.actions        [35660]: NOTICE  [nginx-404] Flush ticket(s) with nftables
2026-10-30 17:26:20,569 fail2ban.actions        [35660]: NOTICE  [sshd] Flush ticket(s) with nftables
2026-10-30 17:26:20,569 fail2ban.jail           [35660]: INFO    Jail 'sshd' stopped
2026-10-30 17:26:20,569 fail2ban.jail           [35660]: INFO    Jail 'nginx-404' stopped
2026-10-30 17:26:20,569 fail2ban.database       [35660]: INFO    Connection to database closed.
2026-10-30 17:26:20,570 fail2ban.server         [35660]: INFO    Exiting Fail2ban
2026-10-30 17:26:20,824 fail2ban.server         [35802]: INFO    ----------------------------------------
--------
2026-10-30 17:26:20,825 fail2ban.server         [35802]: INFO    Starting Fail2ban v1.1.0
2026-10-30 17:26:20,825 fail2ban.observer       [35802]: INFO    Observer start...
2026-10-30 17:26:20,832 fail2ban.database       [35802]: INFO    Connected to fail2ban persistent database
'/var/lib/fail2ban/fail2ban.sqlite3'
2026-10-30 17:26:20,833 fail2ban.jail           [35802]: INFO    Creating new jail 'sshd'
2026-10-30 17:26:20,837 fail2ban.jail           [35802]: INFO    Jail 'sshd' uses systemd {}
2026-10-30 17:26:20,838 fail2ban.jail           [35802]: INFO    Initiated 'systemd' backend
2026-10-30 17:26:20,839 fail2ban.filter         [35802]: INFO      maxLines: 1
2026-10-30 17:26:20,854 fail2ban.filtersystemd  [35802]: INFO    [sshd] Added journal match for: '_SYSTEMD_
UNIT=ssh.service + _COMM=sshd'
2026-10-30 17:26:20,854 fail2ban.filter         [35802]: INFO      maxRetry: 5
2026-10-30 17:26:20,854 fail2ban.filter         [35802]: INFO      findtime: 600
2026-10-30 17:26:20,854 fail2ban.actions        [35802]: INFO      banTime: 3600
2026-10-30 17:26:20,854 fail2ban.filter         [35802]: INFO      encoding: UTF-8
2026-10-30 17:26:20,855 fail2ban.jail           [35802]: INFO    Creating new jail 'nginx-404'
2026-10-30 17:26:20,861 fail2ban.jail           [35802]: INFO    Jail 'nginx-404' uses pyinotify {}
2026-10-30 17:26:20,865 fail2ban.jail           [35802]: INFO    Initiated 'pyinotify' backend
2026-10-30 17:26:20,866 fail2ban.filter         [35802]: INFO      maxRetry: 5
2026-10-30 17:26:20,866 fail2ban.filter         [35802]: INFO      findtime: 600
2026-10-30 17:26:20,866 fail2ban.actions        [35802]: INFO      banTime: 3600
```

```
isabela@isabela-VirtualBox:~$  sudo journalctl -u ssh --since "10 min ago" --no-pager
-- No entries --
isabela@isabela-VirtualBox:~$ sudo tail -n 10 /var/log/nginx/access.log
::1 - - [30/Oct/2026:16:05:11 -0500] "GET /admin/ HTTP/1.1" 401 188 "-" "curl/8.18.0"
::1 - admin [30/Oct/2026:16:05:34 -0500] "GET /admin/ HTTP/1.1" 200 13 "-" "curl/8.18.0"
isabela@isabela-VirtualBox:~$  sudo tail -n 15 /var/log/fail2ban.log
2026-10-30 17:48:36,624 fail2ban.filter         [36340]: INFO      maxRetry: 5
2026-10-30 17:48:36,624 fail2ban.filter         [36340]: INFO      findtime: 600
2026-10-30 17:48:36,624 fail2ban.actions        [36340]: INFO      banTime: 3600
2026-10-30 17:48:36,624 fail2ban.filter         [36340]: INFO      encoding: UTF-8
2026-10-30 17:48:36,624 fail2ban.jail           [36340]: INFO    Creating new jail 'nginx-404'
2026-10-30 17:48:36,703 fail2ban.jail           [36340]: INFO    Jail 'nginx-404' uses pyinotify {}
2026-10-30 17:48:36,707 fail2ban.jail           [36340]: INFO    Initiated 'pyinotify' backend
2026-10-30 17:48:36,708 fail2ban.filter         [36340]: INFO      maxRetry: 5
2026-10-30 17:48:36,708 fail2ban.filter         [36340]: INFO      findtime: 600
2026-10-30 17:48:36,708 fail2ban.actions        [36340]: INFO      banTime: 3600
2026-10-30 17:48:36,708 fail2ban.filter         [36340]: INFO      encoding: UTF-8
2026-10-30 17:48:36,731 fail2ban.filter         [36340]: INFO    Added logfile: '/var/log/nginx/access.log'
 (pos = 175, hash = e8bcca68a7a5bc5ab6dc6cc34d4e2ce930f87971)
2026-10-30 17:48:36,732 fail2ban.filtersystemd  [36340]: INFO    [sshd] Jail is in operation now (process n
ew journal entries)
2026-10-30 17:48:36,742 fail2ban.jail           [36340]: INFO    Jail 'sshd' started
2026-10-30 17:48:36,744 fail2ban.jail           [36340]: INFO    Jail 'nginx-404' started
isabela@isabela-VirtualBox:~$
```

```
2026-10-30 17:26:20,866 fail2ban.actions        [35802]: INFO      banTime: 3600
2026-10-30 17:26:20,866 fail2ban.filter         [35802]: INFO      encoding: UTF-8
2026-10-30 17:26:20,867 fail2ban.filter         [35802]: INFO    Added logfile: '/var/log/nginx/access.log'
 (pos = 175, hash = e8bcca68a7a5bc5ab6dc6cc34d4e2ce930f87971)
2026-10-30 17:26:20,868 fail2ban.filtersystemd  [35802]: INFO    [sshd] Jail is in operation now (process n
ew journal entries)
2026-10-30 17:26:20,869 fail2ban.jail           [35802]: INFO    Jail 'sshd' started
2026-10-30 17:26:20,870 fail2ban.jail           [35802]: INFO    Jail 'nginx-404' started
isabela@isabela-VirtualBox:~$ sudo journalctl -u ssh --since today
oct 30 15:05:24 isabela-VirtualBox systemd[1]: Starting ssh.service - OpenBSD Secure Shell server...
oct 30 15:05:25 isabela-VirtualBox sshd[1617]: Server listening on 0.0.0.0 port 22.
oct 30 15:05:25 isabela-VirtualBox sshd[1617]: Server listening on :: port 22.
oct 30 15:05:25 isabela-VirtualBox systemd[1]: Started ssh.service - OpenBSD Secure Shell server.
oct 30 15:56:01 isabela-VirtualBox systemd[1]: Stopping ssh.service - OpenBSD Secure Shell server...
oct 30 15:56:01 isabela-VirtualBox sshd[1617]: Received signal 15; terminating.
oct 30 15:56:03 isabela-VirtualBox sshd[11720]: Server listening on 0.0.0.0 port 22.
oct 30 15:56:03 isabela-VirtualBox sshd[11720]: Server listening on :: port 22.
oct 30 15:56:31 isabela-VirtualBox sshd[11720]: Received signal 15; terminating.
oct 30 15:56:31 isabela-VirtualBox systemd[1]: Stopping ssh.service - OpenBSD Secure Shell server...
oct 30 15:56:33 isabela-VirtualBox sshd[12312]: Server listening on 0.0.0.0 port 22.
oct 30 15:56:33 isabela-VirtualBox sshd[12312]: Server listening on :: port 22.
oct 30 15:56:33 isabela-VirtualBox systemd[1]: Started ssh.service - OpenBSD Secure Shell server.
isabela@isabela-VirtualBox:~$ sudo tail -n 30 /var/log/nginx/access.log
::1 - - [30/Oct/2026:16:05:11 -0500] "GET /admin/ HTTP/1.1" 401 188 "-" "curl/8.18.0"
::1 - admin [30/Oct/2026:16:05:34 -0500] "GET /admin/ HTTP/1.1" 200 13 "-" "curl/8.18.0"
isabela@isabela-VirtualBox:~$
```

Configuración del registro dedicado y observación de los eventos de seguridad del servidor.

CONFIGUARAR NODO ATACANTE

Preparar el atacante:

```
isabela@isabela-VirtualBox:~$ sudo hostnamectl set-hostname atacante
[sudo: authenticate] Contraseña:
isabela@isabela-VirtualBox:~$ sudo sed -i 's/isabela-VirtualBox/atacante/g' /etc/hosts
isabela@isabela-VirtualBox:~$
```

instalación del sshpass que sirve para optimizar los intentos fallidos. se instala por el adaptador NAT

```
isabela@atacante:~$ sudo apt update && sudo apt install -y sshpass
[sudo: authenticate] Contraseña:
Obj:1 http://archive.ubuntu.com/ubuntu resolute InRelease
Des:2 http://archive.ubuntu.com/ubuntu resolute-updates InRelease [137 kB]
Des:3 http://security.ubuntu.com/ubuntu resolute-security InRelease [137 kB]
Des:4 http://archive.ubuntu.com/ubuntu resolute-backports InRelease [137 kB]
Des:5 http://archive.ubuntu.com/ubuntu resolute-updates/main amd64 Packages [803 kB]
Des:6 http://archive.ubuntu.com/ubuntu resolute-updates/main amd64 Components [110 kB]
Des:7 http://archive.ubuntu.com/ubuntu resolute-updates/universe amd64 Packages [374 kB]
Des:8 http://archive.ubuntu.com/ubuntu resolute-updates/universe amd64 Components [201 kB]
Des:9 http://archive.ubuntu.com/ubuntu resolute-backports/universe amd64 Components [1.056 B]
Des:10 http://security.ubuntu.com/ubuntu resolute-security/main amd64 Components [46,7 kB]
Des:11 http://security.ubuntu.com/ubuntu resolute-security/universe amd64 Components [55,6 kB]
Descargados 2.003 kB en 2s (1.063 kB/s)
Se pueden actualizar 186 paquetes. Ejecute «apt list --upgradable» para verlos.
Instalando:
  sshpass
```

CONEXION DE AMBAS MAQUINAS VIRTUALES:
ambas muestran UP y LOWER_UP significa que están activas y con enlace (el “cable” virtual está conectado)

Servidor:

```
isabela@isabela-VirtualBox:~$ ip -br link
lo               UNKNOWN        00:00:00:00:00:00 <LOOPBACK,UP,LOWER_UP>
enp0s3           UP             08:00:27:8a:1b:8a <BROADCAST,MULTICAST,UP,LOWER_UP>
enp0s8           UP             08:00:27:c2:12:dc <BROADCAST,MULTICAST,UP,LOWER_UP>
isabela@isabela-VirtualBox:~$
```

Atacante:

```
isabela@atacante:~$ ip -br link
lo               UNKNOWN        00:00:00:00:00:00 <LOOPBACK,UP,LOWER_UP>
enp0s3           UP             08:00:27:31:4f:06 <BROADCAST,MULTICAST,UP,LOWER_UP>
enp0s8           UP             08:00:27:21:c4:0e <BROADCAST,MULTICAST,UP,LOWER_UP>
isabela@atacante:~$
```

Servidor:

```
isabela@isabela-VirtualBox:~$ sudo ip addr add 10.10.10.1/24 dev enp0s8
[sudo: authenticate] Contraseña:
isabela@isabela-VirtualBox:~$ sudo ip link set enp0s8 up
isabela@isabela-VirtualBox:~$
```

Atacante:

```
isabela@atacante:~$ sudo ip addr add 10.10.10.2/24 dev enp0s8
[sudo: authenticate] Contraseña:
isabela@atacante:~$ sudo ip link set enp0s8 up
isabela@atacante:~$
```

Servidor:

```
isabela@isabela-VirtualBox:~$ date; ip -4 addr
dom 01 nov 2026 18:05:38 -05
1: lo: <LOOPBACK,UP,LOWER_UP> mtu 65536 qdisc noqueue state UNKNOWN group default qlen 1000
    inet 127.0.0.1/8 scope host lo
       valid_lft forever preferred_lft forever
2: enp0s3: <BROADCAST,MULTICAST,UP,LOWER_UP> mtu 1500 qdisc fq_codel state UP group default qlen 1000
    altname enx0800278a1b8a
    inet 192.168.80.142/24 brd 192.168.80.255 scope global dynamic noprefixroute enp0s3
       valid_lft 41489sec preferred_lft 41489sec
3: enp0s8: <BROADCAST,MULTICAST,UP,LOWER_UP> mtu 1500 qdisc fq_codel state UP group default qlen 1000
    altname enx080027c212dc
    inet 10.10.10.1/24 scope global enp0s8
       valid_lft forever preferred_lft forever
isabela@isabela-VirtualBox:~$
```

Atacante:

```
isabela@atacante:~$ date; ip -4 addr
dom 01 nov 2026 18:03:12 -05
1: lo: <LOOPBACK,UP,LOWER_UP> mtu 65536 qdisc noqueue state UNKNOWN group default qlen 1000
    inet 127.0.0.1/8 scope host lo
       valid_lft forever preferred_lft forever
2: enp0s3: <BROADCAST,MULTICAST,UP,LOWER_UP> mtu 1500 qdisc fq_codel state UP group default qlen 1000
    altname enx080027314f06
    inet 10.0.2.15/24 brd 10.0.2.255 scope global dynamic noprefixroute enp0s3
       valid_lft 85694sec preferred_lft 85694sec
3: enp0s8: <BROADCAST,MULTICAST,UP,LOWER_UP> mtu 1500 qdisc fq_codel state UP group default qlen 1000
    altname enx08002721c40e
    inet 10.10.10.2/24 scope global enp0s8
       valid_lft forever preferred_lft forever
isabela@atacante:~$
```

comprobacion:

```
isabela@atacante:~$ ping -c 3 10.10.10.1
PING 10.10.10.1 (10.10.10.1) 56(84) bytes of data.
64 bytes from 10.10.10.1: icmp_seq=1 ttl=64 time=0.403 ms
64 bytes from 10.10.10.1: icmp_seq=2 ttl=64 time=0.484 ms
64 bytes from 10.10.10.1: icmp_seq=3 ttl=64 time=0.367 ms

--- 10.10.10.1 ping statistics ---
3 packets transmitted, 3 received, 0% packet loss, time 2049ms
rtt min/avg/max/mdev = 0.367/0.418/0.484/0.048 ms
isabela@atacante:~$
```

paso 14:

Atacante:

```
isabela@atacante:~$ ssh usuario_prueba@10.10.10.1
The authenticity of host '10.10.10.1 (10.10.10.1)' can't be established.
ED25519 key fingerprint is: SHA256:qhajO6575nIhwkWSgIt7nGm+9ZgpheyvH8lmjQPCor8
This host key is known by the following other names/addresses:
    ~/.ssh/known_hosts:1: [hashed name]
Are you sure you want to continue connecting (yes/no/[fingerprint])? yes
Warning: Permanently added '10.10.10.1' (ED25519) to the list of known hosts.
usuario_prueba@10.10.10.1's password:
Permission denied, please try again.
usuario_prueba@10.10.10.1's password:
Connection closed by 10.10.10.1 port 22
isabela@atacante:~$
```

```
isabela@atacante:~$ ssh usuario_prueba@10.10.10.1
usuario_prueba@10.10.10.1's password:
Permission denied, please try again.
usuario_prueba@10.10.10.1's password:
Permission denied, please try again.
usuario_prueba@10.10.10.1's password:
Received disconnect from 10.10.10.1 port 22:2: Too many authentication failures
Disconnected from 10.10.10.1 port 22
isabela@atacante:~$
```

Servidor:
Comprobar el bloqueo

```
isabela@isabela-VirtualBox:~$ sudo fail2ban-client status sshd
[sudo: authenticate] Contraseña:
Status for the jail: sshd
|- Filter
|  |- Currently failed: 1
|  |- Total failed:     6
|  `- Journal matches:  _SYSTEMD_UNIT=ssh.service + _COMM=sshd
`- Actions
   |- Currently banned: 1
   |- Total banned:     1
   `- Banned IP list:   10.10.10.2
isabela@isabela-VirtualBox:~$
```

```
isabela@isabela-VirtualBox:~$ sudo grep -E "Found|Ban" /var/log/fail2ban.log
2026-11-01 18:11:34,369 fail2ban.filter         [1570]: INFO    [sshd] Found 10.10.10.2 - 2026-11-01 18:11:33
2026-11-01 18:11:39,655 fail2ban.filter         [1570]: INFO    [sshd] Found 10.10.10.2 - 2026-11-01 18:11:39
2026-11-01 18:14:48,405 fail2ban.filter         [1570]: INFO    [sshd] Found 10.10.10.2 - 2026-11-01 18:14:47
2026-11-01 18:14:52,155 fail2ban.filter         [1570]: INFO    [sshd] Found 10.10.10.2 - 2026-11-01 18:14:51
2026-11-01 18:14:55,532 fail2ban.filter         [1570]: INFO    [sshd] Found 10.10.10.2 - 2026-11-01 18:14:55
2026-11-01 18:14:55,532 fail2ban.filter         [1570]: INFO    [sshd] Found 10.10.10.2 - 2026-11-01 18:14:55
2026-11-01 18:14:55,599 fail2ban.actions        [1570]: NOTICE  [sshd] Ban 10.10.10.2
isabela@isabela-VirtualBox:~$ sudo journalctl -u ssh --since "10 min ago" --no-pager | grep -i "failed"
nov 01 18:11:33 isabela-VirtualBox sshd-session[5722]: Failed password for usuario_prueba from 10.10.10.2 port 5
3642 ssh2
nov 01 18:11:37 isabela-VirtualBox unix_chkpwd[5731]: password check failed for user (usuario_prueba)
nov 01 18:11:39 isabela-VirtualBox sshd-session[5722]: Failed password for usuario_prueba from 10.10.10.2 port 5
3642 ssh2
nov 01 18:14:46 isabela-VirtualBox unix_chkpwd[5750]: password check failed for user (usuario_prueba)
nov 01 18:14:47 isabela-VirtualBox sshd-session[5748]: Failed password for usuario_prueba from 10.10.10.2 port 4
5192 ssh2
nov 01 18:14:51 isabela-VirtualBox sshd-session[5748]: Failed password for usuario_prueba from 10.10.10.2 port 4
5192 ssh2
nov 01 18:14:53 isabela-VirtualBox unix_chkpwd[5752]: password check failed for user (usuario_prueba)
nov 01 18:14:55 isabela-VirtualBox sshd-session[5748]: Failed password for usuario_prueba from 10.10.10.2 port 4
5192 ssh2
isabela@isabela-VirtualBox:~$
```

PRUEBA DE QUE EL ATACANTE QUEDÓ BLOQUEADO

```
isabela@atacante:~$ ssh -o ConnectTimeout=5 usuario_prueba@10.10.10.1
ssh: connect to host 10.10.10.1 port 22: Connection refused
isabela@atacante:~$
```

Detección y bloqueo automático de la dirección IP del nodo atacante después de superar el umbral de intentos fallidos.

Paso 16. Probar la segunda jail
servidor:

```
isabela@isabela-VirtualBox:~$ date; sudo fail2ban-client status nginx-404
dom 01 nov 2026 23:06:25 -05
[sudo: authenticate] Contraseña:
Status for the jail: nginx-404
|- Filter
|  |- Currently failed: 0
|  |- Total failed:     0
|  `- File list:        /var/log/nginx/access.log
`- Actions
   |- Currently banned: 0
   |- Total banned:     0
   `- Banned IP list:
isabela@isabela-VirtualBox:~$
```

Atacante:

```
isabela@atacante:~$ for i in {1..6}; do curl -s -o /dev/null -w "%{http_code}\n" --max-time 5 -u admin:mala
 http://10.10.10.1/admin/;done
401
401
401
401
401
401
isabela@atacante:~$
```

Servidor:

```
isabela@isabela-VirtualBox:~$ sudo tail -n 20 /var/log/nginx/access.log
10.10.10.2 - admin [01/Nov/2026:23:09:33 -0500] "GET /admin/ HTTP/1.1" 401 188 "-" "curl/8.18.0"
10.10.10.2 - admin [01/Nov/2026:23:09:33 -0500] "GET /admin/ HTTP/1.1" 401 188 "-" "curl/8.18.0"
10.10.10.2 - admin [01/Nov/2026:23:09:33 -0500] "GET /admin/ HTTP/1.1" 401 188 "-" "curl/8.18.0"
10.10.10.2 - admin [01/Nov/2026:23:09:33 -0500] "GET /admin/ HTTP/1.1" 401 188 "-" "curl/8.18.0"
10.10.10.2 - admin [01/Nov/2026:23:09:33 -0500] "GET /admin/ HTTP/1.1" 401 188 "-" "curl/8.18.0"
10.10.10.2 - admin [01/Nov/2026:23:09:33 -0500] "GET /admin/ HTTP/1.1" 401 188 "-" "curl/8.18.0"
isabela@isabela-VirtualBox:~$ sudo tail -n 5 /var/log/nginx/error.log
2026/11/01 23:09:33 [error] 3039#3039: *2 user "admin": password mismatch, client: 10.10.10.2, server: _, r
equest: "GET /admin/ HTTP/1.1", host: "10.10.10.1"
2026/11/01 23:09:33 [error] 3039#3039: *3 user "admin": password mismatch, client: 10.10.10.2, server: _, r
equest: "GET /admin/ HTTP/1.1", host: "10.10.10.1"
2026/11/01 23:09:33 [error] 3039#3039: *4 user "admin": password mismatch, client: 10.10.10.2, server: _, r
equest: "GET /admin/ HTTP/1.1", host: "10.10.10.1"
2026/11/01 23:09:33 [error] 3039#3039: *5 user "admin": password mismatch, client: 10.10.10.2, server: _, r
equest: "GET /admin/ HTTP/1.1", host: "10.10.10.1"
2026/11/01 23:09:33 [error] 3039#3039: *6 user "admin": password mismatch, client: 10.10.10.2, server: _, r
equest: "GET /admin/ HTTP/1.1", host: "10.10.10.1"
```

```
isabela@isabela-VirtualBox:~$ sudo fail2ban-regex /var/log/nginx/access.log nginx-404

Running tests
=============

Use            jail : nginx-404
Use         log file : /var/log/nginx/access.log
Use         encoding : UTF-8


Results
=======

Failregex: 6 total
|-  #) [# of hits] regular expression
|   1) [6] ^<HOST> - .* "(GET|POST|HEAD) /admin[^"]*" (401|403|404) .*
`-

Ignoreregex: 0 total

Date template hits:
|- [# of hits] date format
|  [6] Day(?P<_sep>[-/])MON(?P=_sep)ExYear[ :]?24hour:Minute:Second(?:\.Microseconds)?(?: Zone offset)?
`-

Lines: 6 lines, 0 ignored, 6 matched, 0 missed
[processed in 0.02 sec]
```

```
isabela@isabela-VirtualBox:~$ sudo fail2ban-client status nginx-404
Status for the jail: nginx-404
|- Filter
|  |- Currently failed: 1
|  |- Total failed:     6
|  `- File list:        /var/log/nginx/access.log
`- Actions
   |- Currently banned: 1
   |- Total banned:     1
   `- Banned IP list:   10.10.10.2
isabela@isabela-VirtualBox:~$ sudo grep -E "Found|Ban" /var/log/fail2ban.log | grep nginx-404
2026-11-01 23:09:33,648 fail2ban.filter         [1570]: INFO    [nginx-404] Found 10.10.10.2 - 2026-11-01 2
3:09:33
2026-11-01 23:09:33,648 fail2ban.filter         [1570]: INFO    [nginx-404] Found 10.10.10.2 - 2026-11-01 2
3:09:33
2026-11-01 23:09:33,649 fail2ban.filter         [1570]: INFO    [nginx-404] Found 10.10.10.2 - 2026-11-01 2
3:09:33
2026-11-01 23:09:33,649 fail2ban.filter         [1570]: INFO    [nginx-404] Found 10.10.10.2 - 2026-11-01 2
3:09:33
2026-11-01 23:09:33,649 fail2ban.filter         [1570]: INFO    [nginx-404] Found 10.10.10.2 - 2026-11-01 2
3:09:33
2026-11-01 23:09:33,650 fail2ban.filter         [1570]: INFO    [nginx-404] Found 10.10.10.2 - 2026-11-01 2
3:09:33
2026-11-01 23:09:33,822 fail2ban.actions        [1570]: NOTICE  [nginx-404] Ban 10.10.10.2
```

```
isabela@isabela-VirtualBox:~$ sudo nft list ruleset | grep -B2 -A6 f2b
# Warning: table ip filter is managed by iptables-nft, do not touch!
# Warning: table ip6 filter is managed by iptables-nft, do not touch!
        }
}
table inet f2b-table {
        set addr-set-sshd {
                type ipv4_addr
        }

        set addr-set-nginx-404 {
                type ipv4_addr
--
        }

        chain f2b-chain {
                type filter hook input priority filter - 1; policy accept;
                tcp dport 22 ip saddr @addr-set-sshd reject with icmp port-unreachable
                tcp dport { 80, 443 } ip saddr @addr-set-nginx-404 reject with icmp port-unreachable
        }
}
```

PASO 17:
Servidor:

```
isabela@isabela-VirtualBox:~$ date; sudo fail2ban-client banned
dom 01 nov 2026 23:29:02 -05
[sudo: authenticate] Contraseña:
[{'sshd': []}, {'nginx-404': ['10.10.10.2']}]
isabela@isabela-VirtualBox:~$ sudo fail2ban-client unban 10.10.10.2
1
isabela@isabela-VirtualBox:~$ sudo nano /etc/fail2ban/jail.local
isabela@isabela-VirtualBox:~$
```

```
  GNU nano 8.7.1                                  /etc/fail2ban/jail.local *
[DEFAULT]
bantime = 1h
findtime = 10m
maxretry = 5
ignoreip = 127.0.0.1/8 ::1 10.10.10.2

[sshd]
enabled = true
backend = systemd
port = ssh
maxretry = 5
findtime = 10m
bantime = 1h

[nginx-404]
enabled = true
filter = nginx-404
port = http,https
logpath = /var/log/nginx/access.log
maxretry = 5
findtime = 10m
bantime = 1h
```

```
isabela@isabela-VirtualBox:~$ sudo fail2ban-client -t
OK: configuration test is successful
isabela@isabela-VirtualBox:~$ sudo systemctl restart fail2ban
isabela@isabela-VirtualBox:~$ sudo fail2ban-client get sshd ignoreip
These IP addresses/networks are ignored:
|- 127.0.0.0/8
|- ::1
`- 10.10.10.2
isabela@isabela-VirtualBox:~$ sudo fail2ban-client get nginx-404 ignoreip
These IP addresses/networks are ignored:
|- 127.0.0.0/8
|- ::1
`- 10.10.10.2
isabela@isabela-VirtualBox:~$
```

Atacante:

```
isabela@atacante:~$ for i in {1..6}; do sshpass -p mala ssh -o StrictHostKeyChecking=no -o PubkeyAuthentica
tion=no -o ConnectTimeout=5 usuario_prueba@10.10.10.1; done
Permission denied, please try again.
Permission denied, please try again.
Permission denied, please try again.
Permission denied, please try again.
Permission denied, please try again.
Permission denied, please try again.
isabela@atacante:~$ for i in {1..8}; do curl -s -o /dev/null -w "%{http_code}\n" --max-time 5 -u admin:mala
 http://10.10.10.1/admin/; sleep 1; done
401
401
401
401
401
401
401
401
isabela@atacante:~$
```

Comprobar que sigue entrando

```
usuario_prueba@isabela-VirtualBox: ~ — ssh -o ConnectTimeout=5 usuario_prueba...

isabela@atacante:~$ ssh -o ConnectTimeout=5 usuario_prueba@10.10.10.1
usuario_prueba@10.10.10.1's password:
Welcome to Ubuntu 26.04 LTS (GNU/Linux 7.0.0-38-generic x86_64)

 * Documentation:  https://docs.ubuntu.com
 * Management:     https://landscape.canonical.com
 * Support:        https://ubuntu.com/pro

El mantenimiento de seguridad expandido para Applications está desactivado

Se pueden aplicar 194 actualizaciones de forma inmediata.
Para ver estas actualizaciones adicionales, ejecute: apt list --upgradable

Active ESM Apps para recibir futuras actualizaciones de seguridad adicionales.
Vea https://ubuntu.com/esm o ejecute «sudo pro status»


The programs included with the Ubuntu system are free software;
the exact distribution terms for each program are described in the
individual files in /usr/share/doc/*/copyright.

Ubuntu comes with ABSOLUTELY NO WARRANTY, to the extent permitted by
applicable law.

usuario_prueba@isabela-VirtualBox:~$
```

Servidor:

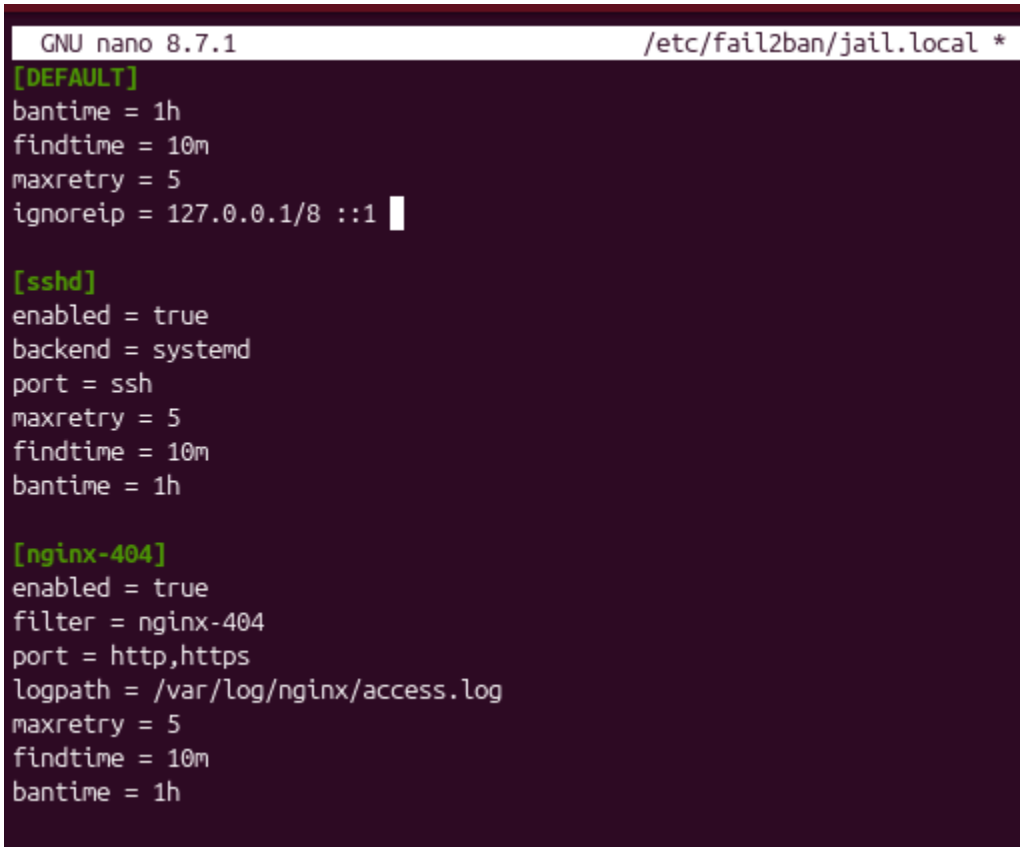

isabela@isabela-VirtualBox:~$ sudo fail2ban-client status sshd
Status for the jail: sshd
|- Filter
| |- Currently failed: 0
| |- Total failed: 0
| `- Journal matches: _SYSTEMD_UNIT=ssh.service + _COMM=sshd
`- Actions
   |- Currently banned: 0
   |- Total banned: 0
   `- Banned IP list:
isabela@isabela-VirtualBox:~$ sudo fail2ban-client status nginx-404
Status for the jail: nginx-404
|- Filter
| |- Currently failed: 0
| |- Total failed: 0
| `- File list: /var/log/nginx/access.log
`- Actions
   |- Currently banned: 0
   |- Total banned: 0
   `- Banned IP list:

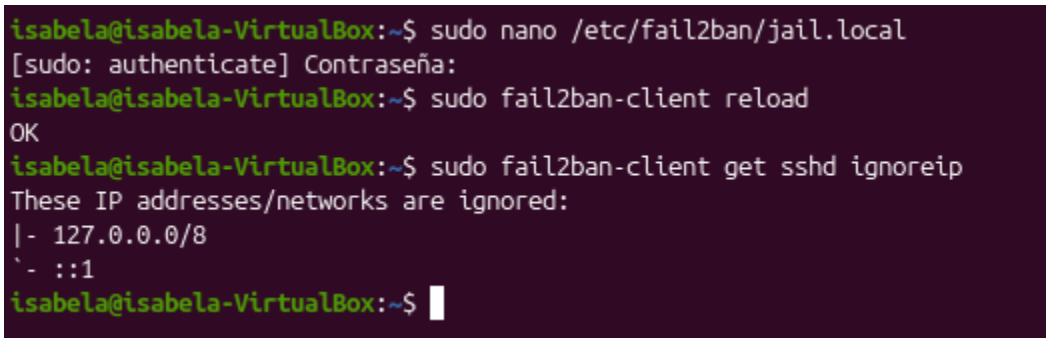
```
    - Banned IP list:
isabela@isabela-VirtualBox:~$ sudo grep -i "ignore" /var/log/fail2ban.log | tail
2026-11-01 23:37:07,905 fail2ban.filter         [9167]: INFO    [sshd] Ignore 10.10.10.2 by ip
2026-11-01 23:37:09,905 fail2ban.filter         [9167]: INFO    [sshd] Ignore 10.10.10.2 by ip
2026-11-01 23:37:30,038 fail2ban.filter         [9167]: INFO    [nginx-404] Ignore 10.10.10.2 by ip
2026-11-01 23:37:31,185 fail2ban.filter         [9167]: INFO    [nginx-404] Ignore 10.10.10.2 by ip
2026-11-01 23:37:32,223 fail2ban.filter         [9167]: INFO    [nginx-404] Ignore 10.10.10.2 by ip
2026-11-01 23:37:33,251 fail2ban.filter         [9167]: INFO    [nginx-404] Ignore 10.10.10.2 by ip
2026-11-01 23:37:34,276 fail2ban.filter         [9167]: INFO    [nginx-404] Ignore 10.10.10.2 by ip
2026-11-01 23:37:35,296 fail2ban.filter         [9167]: INFO    [nginx-404] Ignore 10.10.10.2 by ip
2026-11-01 23:37:36,311 fail2ban.filter         [9167]: INFO    [nginx-404] Ignore 10.10.10.2 by ip
2026-11-01 23:37:37,362 fail2ban.filter         [9167]: INFO    [nginx-404] Ignore 10.10.10.2 by ip
isabela@isabela-VirtualBox:~$ sudo journalctl -u ssh --since "10 min ago" --no-pager | grep "10.10.10.2"
nov 01 23:36:55 isabela-VirtualBox sshd-session[9219]: pam_unix(sshd:auth): authentication failure; logname
= uid=0 euid=0 tty=ssh ruser= rhost=10.10.10.2  user=usuario_prueba
nov 01 23:36:57 isabela-VirtualBox sshd-session[9219]: Failed password for usuario_prueba from 10.10.10.2 p
ort 43826 ssh2
nov 01 23:36:58 isabela-VirtualBox sshd-session[9219]: Connection closed by authenticating user usuario_pru
eba 10.10.10.2 port 43826 [preauth]
nov 01 23:36:58 isabela-VirtualBox sshd-session[9223]: pam_unix(sshd:auth): authentication failure; logname
= uid=0 euid=0 tty=ssh ruser= rhost=10.10.10.2  user=usuario_prueba
nov 01 23:36:59 isabela-VirtualBox sshd-session[9223]: Failed password for usuario_prueba from 10.10.10.2 p
ort 43830 ssh2
nov 01 23:37:00 isabela-VirtualBox sshd-session[9223]: Connection closed by authenticating user usuario_pru
eba 10.10.10.2 port 43830 [preauth]
nov 01 23:37:00 isabela-VirtualBox sshd-session[9228]: pam_unix(sshd:auth): authentication failure; logname
= uid=0 euid=0 tty=ssh ruser= rhost=10.10.10.2  user=usuario_prueba
nov 01 23:37:02 isabela-VirtualBox sshd-session[9228]: Failed password for usuario_prueba from 10.10.10.2 p
ort 35922 ssh2
nov 01 23:37:02 isabela-VirtualBox sshd-session[9228]: Connection closed by authenticating user usuario_pru
eba 10.10.10.2 port 35922 [preauth]
```

```
eba 10.10.10.2 port 35936 [preauth]
nov 01 23:37:05 isabela-VirtualBox sshd-session[9236]: pam_unix(sshd:auth): authentication failure; logname
= uid=0 euid=0 tty=ssh ruser= rhost=10.10.10.2  user=usuario_prueba
nov 01 23:37:07 isabela-VirtualBox sshd-session[9236]: Failed password for usuario_prueba from 10.10.10.2 p
ort 35940 ssh2
nov 01 23:37:07 isabela-VirtualBox sshd-session[9236]: Connection closed by authenticating user usuario_pru
eba 10.10.10.2 port 35940 [preauth]
nov 01 23:37:07 isabela-VirtualBox sshd-session[9248]: pam_unix(sshd:auth): authentication failure; logname
= uid=0 euid=0 tty=ssh ruser= rhost=10.10.10.2  user=usuario_prueba
nov 01 23:37:09 isabela-VirtualBox sshd-session[9248]: Failed password for usuario_prueba from 10.10.10.2 p
ort 35956 ssh2
nov 01 23:37:10 isabela-VirtualBox sshd-session[9248]: Connection closed by authenticating user usuario_pru
eba 10.10.10.2 port 35956 [preauth]
nov 01 23:37:10 isabela-VirtualBox sshd[1602]: srclimit_penalise: 10.10.10.2/32: activating ipv4 penalty of
 18 seconds for penalty: failed authentication
nov 01 23:38:35 isabela-VirtualBox sshd-session[9262]: Accepted password for usuario_prueba from 10.10.10.2
 port 44990 ssh2
isabela@isabela-VirtualBox:~$ sudo tail -n 10 /var/log/nginx/access.log
10.10.10.2 - admin [01/Nov/2026:23:09:33 -0500] "GET /admin/ HTTP/1.1" 401 188 "-" "curl/8.18.0"
10.10.10.2 - admin [01/Nov/2026:23:09:33 -0500] "GET /admin/ HTTP/1.1" 401 188 "-" "curl/8.18.0"
10.10.10.2 - admin [01/Nov/2026:23:37:30 -0500] "GET /admin/ HTTP/1.1" 401 188 "-" "curl/8.18.0"
10.10.10.2 - admin [01/Nov/2026:23:37:31 -0500] "GET /admin/ HTTP/1.1" 401 188 "-" "curl/8.18.0"
10.10.10.2 - admin [01/Nov/2026:23:37:32 -0500] "GET /admin/ HTTP/1.1" 401 188 "-" "curl/8.18.0"
10.10.10.2 - admin [01/Nov/2026:23:37:33 -0500] "GET /admin/ HTTP/1.1" 401 188 "-" "curl/8.18.0"
10.10.10.2 - admin [01/Nov/2026:23:37:34 -0500] "GET /admin/ HTTP/1.1" 401 188 "-" "curl/8.18.0"
10.10.10.2 - admin [01/Nov/2026:23:37:35 -0500] "GET /admin/ HTTP/1.1" 401 188 "-" "curl/8.18.0"
10.10.10.2 - admin [01/Nov/2026:23:37:36 -0500] "GET /admin/ HTTP/1.1" 401 188 "-" "curl/8.18.0"
10.10.10.2 - admin [01/Nov/2026:23:37:37 -0500] "GET /admin/ HTTP/1.1" 401 188 "-" "curl/8.18.0"
isabela@isabela-VirtualBox:~$
```

Simular un incidente y solucionarlo
PASO 18:

primero quitamos la ip de whiteList para que en este ejercicio salga bloqueado





simular el bloqueo

```
isabela@atacante:~$ for i in {1..6}; do sshpass -p mala ssh -o StrictHostKeyChecking=no -o PubkeyAuthentica
tion=no -o ConnectTimeout=5 usuario_prueba@10.10.10.1; done
Permission denied, please try again.
Permission denied, please try again.
Permission denied, please try again.
Permission denied, please try again.
Permission denied, please try again.
kex_exchange_identification: read: Connection reset by peer
Connection reset by 10.10.10.1 port 22
isabela@atacante:~$ ssh -o ConnectTimeout=5 usuario_prueba@10.10.10.1
ssh: connect to host 10.10.10.1 port 22: Connection refused
isabela@atacante:~$ ping -c 3 10.10.10.1
PING 10.10.10.1 (10.10.10.1) 56(84) bytes of data.
64 bytes from 10.10.10.1: icmp_seq=1 ttl=64 time=0.443 ms
64 bytes from 10.10.10.1: icmp_seq=2 ttl=64 time=0.533 ms
64 bytes from 10.10.10.1: icmp_seq=3 ttl=64 time=0.504 ms

--- 10.10.10.1 ping statistics ---
3 packets transmitted, 3 received, 0% packet loss, time 2042ms
rtt min/avg/max/mdev = 0.443/0.493/0.533/0.037 ms
isabela@atacante:~$
```

```
isabela@isabela-VirtualBox:~$ systemctl is-active ssh
active
isabela@isabela-VirtualBox:~$ sudo ufw status | grep -E "22|OpenSSH"
[sudo: authenticate] Contraseña:
2224/tcp                   ALLOW       Anywhere
OpenSSH                    ALLOW       Anywhere
22/tcp                     ALLOW       Anywhere
2224/tcp (v6)              ALLOW       Anywhere (v6)
OpenSSH (v6)               ALLOW       Anywhere (v6)
22/tcp (v6)                ALLOW       Anywhere (v6)
isabela@isabela-VirtualBox:~$ sudo fail2ban-client status sshd
Status for the jail: sshd
|- Filter
|  |- Currently failed: 0
|  |- Total failed:     5
|  `- Journal matches:  _SYSTEMD_UNIT=ssh.service + _COMM=sshd
`- Actions
   |- Currently banned: 1
   |- Total banned:     1
   `- Banned IP list:   10.10.10.2
isabela@isabela-VirtualBox:~$ sudo grep -E "Found|Ban|Unban" /var/log/fail2ban.log | grep 10.10.10.2
2026-11-01 18:11:34,369 fail2ban.filter         [1570]: INFO    [sshd] Found 10.10.10.2 - 2026-11-01 18:11:
33
2026-11-01 18:11:39,655 fail2ban.filter         [1570]: INFO    [sshd] Found 10.10.10.2 - 2026-11-01 18:11:
39
2026-11-01 18:14:48,405 fail2ban.filter         [1570]: INFO    [sshd] Found 10.10.10.2 - 2026-11-01 18:14:
47
2026-11-01 18:14:52,155 fail2ban.filter         [1570]: INFO    [sshd] Found 10.10.10.2 - 2026-11-01 18:14:
51
2026-11-01 18:14:55,532 fail2ban.filter         [1570]: INFO    [sshd] Found 10.10.10.2 - 2026-11-01 18:14:
55
2026-11-01 18:14:55,532 fail2ban.filter         [1570]: INFO    [sshd] Found 10.10.10.2 - 2026-11-01 18:14:
55
```

```
2026-11-01 23:09:33,822 fail2ban.actions        [1570]: NOTICE  [nginx-404] Ban 10.10.10.2
2026-11-01 23:29:46,628 fail2ban.actions        [1570]: NOTICE  [nginx-404] Unban 10.10.10.2
2026-11-02 20:33:19,584 fail2ban.filter         [1562]: INFO    [sshd] Found 10.10.10.2 - 2026-11-02 20:33:
18
2026-11-02 20:33:20,846 fail2ban.filter         [1562]: INFO    [sshd] Found 10.10.10.2 - 2026-11-02 20:33:
20
2026-11-02 20:33:22,846 fail2ban.filter         [1562]: INFO    [sshd] Found 10.10.10.2 - 2026-11-02 20:33:
22
2026-11-02 20:33:26,096 fail2ban.filter         [1562]: INFO    [sshd] Found 10.10.10.2 - 2026-11-02 20:33:
25
2026-11-02 20:33:28,640 fail2ban.filter         [1562]: INFO    [sshd] Found 10.10.10.2 - 2026-11-02 20:33:
28
2026-11-02 20:33:28,754 fail2ban.actions        [1562]: NOTICE  [sshd] Ban 10.10.10.2
isabela@isabela-VirtualBox:~$ sudo journalctl -u ssh --since "15 min ago" --no-pager | grep "Failed" | grep
 10.10.10.2
nov 02 20:33:18 isabela-VirtualBox sshd-session[28654]: Failed password for usuario_prueba from 10.10.10.2
port 50034 ssh2
nov 02 20:33:20 isabela-VirtualBox sshd-session[28657]: Failed password for usuario_prueba from 10.10.10.2
port 50050 ssh2
nov 02 20:33:22 isabela-VirtualBox sshd-session[28663]: Failed password for usuario_prueba from 10.10.10.2
port 50062 ssh2
nov 02 20:33:25 isabela-VirtualBox sshd-session[28666]: Failed password for usuario_prueba from 10.10.10.2
port 50066 ssh2
nov 02 20:33:28 isabela-VirtualBox sshd-session[28669]: Failed password for usuario_prueba from 10.10.10.2
port 35130 ssh2
isabela@isabela-VirtualBox:~$ sudo fail2ban-client get sshd maxretry
5
isabela@isabela-VirtualBox:~$ sudo fail2ban-client get sshd findtime
600
isabela@isabela-VirtualBox:~$ sudo fail2ban-client get sshd bantime
3600
isabela@isabela-VirtualBox:~$
```

desbloquear sin apagar Fail2Ban

```
isabela@isabela-VirtualBox:~$ sudo fail2ban-client set sshd unbanip 10.10.10.2
[sudo: authenticate] Contraseña:
1
isabela@isabela-VirtualBox:~$ sudo fail2ban-client status sshd
Status for the jail: sshd
|- Filter
|  |- Currently failed: 0
|  |- Total failed:     5
|  `- Journal matches:  _SYSTEMD_UNIT=ssh.service + _COMM=sshd
`- Actions
   |- Currently banned: 0
   |- Total banned:     1
   `- Banned IP list:
isabela@isabela-VirtualBox:~$ sudo grep Unban /var/log/fail2ban.log
2026-11-01 19:14:55,899 fail2ban.actions        [1570]: NOTICE  [sshd] Unban 10.10.10.2
2026-11-01 23:29:46,628 fail2ban.actions        [1570]: NOTICE  [nginx-404] Unban 10.10.10.2
2026-11-02 20:38:13,530 fail2ban.actions        [1562]: NOTICE  [sshd] Unban 10.10.10.2
isabela@isabela-VirtualBox:~$ systemctl is-active fail2ban
active
isabela@isabela-VirtualBox:~$ sudo fail2ban-client status
Status
|- Number of jail:      2
`- Jail list:   nginx-404, sshd
isabela@isabela-VirtualBox:~$
```

```
isabela@atacante:~$ ssh usuario_prueba@10.10.10.1
usuario_prueba@10.10.10.1's password:
Last login: Sun Nov  1 23:38:38 -05 2026 from 10.10.10.2 on ssh
Welcome to Ubuntu 26.04 LTS (GNU/Linux 7.0.0-38-generic x86_64)

 * Documentation:  https://docs.ubuntu.com
 * Management:     https://landscape.canonical.com
 * Support:        https://ubuntu.com/pro

El mantenimiento de seguridad expandido para Applications está desactivado

Se pueden aplicar 178 actualizaciones de forma inmediata.
Para ver estas actualizaciones adicionales, ejecute: apt list --upgradable

Active ESM Apps para recibir futuras actualizaciones de seguridad adicionales.
Vea https://ubuntu.com/esm o ejecute «sudo pro status»

*** Es necesario reiniciar el sistema ***
Last login: Sun Nov  1 23:38:42 2026 from 10.10.10.2
usuario_prueba@isabela-VirtualBox:~$
```

```
  GNU nano 8.7.1                                /etc/fail2ban/jail.local *
[DEFAULT]
bantime = 1h
findtime = 10m
maxretry = 5
ignoreip = 127.0.0.1/8 ::1 10.10.10.2

[sshd]
enabled = true
backend = systemd
port = ssh
maxretry = 5
findtime = 10m
bantime = 1h

[nginx-404]
enabled = true
filter = nginx-404
port = http,https
logpath = /var/log/nginx/access.log
maxretry = 5
findtime = 10m
bantime = 1h
```

```
isabela@isabela-VirtualBox:~$ sudo nano /etc/fail2ban/jail.local
isabela@isabela-VirtualBox:~$ sudo fail2ban-client reload
OK
isabela@isabela-VirtualBox:~$ sudo fail2ban-client get sshd ignoreip
These IP addresses/networks are ignored:
|- 127.0.0.0/8
|- ::1
`- 10.10.10.2
isabela@isabela-VirtualBox:~$
```

verificación final

```
isabela@isabela-VirtualBox:~$ date
lun 02 nov 2026 20:46:31 -05
isabela@isabela-VirtualBox:~$ sudo fail2ban-client status
Status
|- Number of jail:      2
`- Jail list:   nginx-404, sshd
isabela@isabela-VirtualBox:~$ sudo fail2ban-client status nginx-404
Status for the jail: nginx-404
|- Filter
|  |- Currently failed: 0
|  |- Total failed:     0
|  `- File list:        /var/log/nginx/access.log
`- Actions
   |- Currently banned: 0
   |- Total banned:     0
   `- Banned IP list:
isabela@isabela-VirtualBox:~$ sudo fail2ban-client status sshd
Status for the jail: sshd
|- Filter
|  |- Currently failed: 0
|  |- Total failed:     5
|  `- Journal matches:  _SYSTEMD_UNIT=ssh.service + _COMM=sshd
`- Actions
   |- Currently banned: 0
   |- Total banned:     1
   `- Banned IP list:
isabela@isabela-VirtualBox:~$ sudo fail2ban-client get sshd ignoreip
These IP addresses/networks are ignored:
|- 127.0.0.0/8
|- ::1
`- 10.10.10.2
```

```
isabela@isabela-VirtualBox:~$ sudo fail2ban-client get nginx-404 ignoreip
These IP addresses/networks are ignored:
|- 127.0.0.0/8
|- ::1
`- 10.10.10.2
isabela@isabela-VirtualBox:~$ sudo tail -n 50 /var/log/fail2ban.log
2026-11-02 16:18:25,719 fail2ban.filter         [1562]: INFO      findtime: 600
2026-11-02 16:18:25,719 fail2ban.actions        [1562]: INFO      banTime: 3600
2026-11-02 16:18:25,719 fail2ban.filter         [1562]: INFO      encoding: UTF-8
2026-11-02 16:18:25,725 fail2ban.server         [1562]: INFO    Jail 'sshd' reloaded
2026-11-02 16:18:25,725 fail2ban.server         [1562]: INFO    Jail 'nginx-404' reloaded
2026-11-02 16:18:25,725 fail2ban.server         [1562]: INFO    Reload finished.
2026-11-02 19:46:25,610 fail2ban.server         [1562]: INFO    Reload all jails
2026-11-02 19:46:25,610 fail2ban.filtersystemd  [1562]: INFO    [sshd] Removed journal match for: '*'
2026-11-02 19:46:25,610 fail2ban.server         [1562]: INFO    Reload jail 'sshd'
2026-11-02 19:46:25,610 fail2ban.filter         [1562]: INFO      maxLines: 1
2026-11-02 19:46:25,612 fail2ban.filtersystemd  [1562]: INFO    [sshd] Added journal match for: '_SYSTEMD_U
NIT=ssh.service + _COMM=sshd'
2026-11-02 19:46:25,614 fail2ban.filter         [1562]: INFO      maxRetry: 5
2026-11-02 19:46:25,615 fail2ban.filter         [1562]: INFO      findtime: 600
2026-11-02 19:46:25,615 fail2ban.actions        [1562]: INFO      banTime: 3600
2026-11-02 19:46:25,615 fail2ban.filter         [1562]: INFO      encoding: UTF-8
2026-11-02 19:46:25,615 fail2ban.server         [1562]: INFO    Reload jail 'nginx-404'
2026-11-02 19:46:25,615 fail2ban.filter         [1562]: INFO      maxRetry: 5
2026-11-02 19:46:25,615 fail2ban.filter         [1562]: INFO      findtime: 600
2026-11-02 19:46:25,616 fail2ban.actions        [1562]: INFO      banTime: 3600
2026-11-02 19:46:25,616 fail2ban.filter         [1562]: INFO      encoding: UTF-8
2026-11-02 19:46:25,617 fail2ban.server         [1562]: INFO    Jail 'sshd' reloaded
2026-11-02 19:46:25,617 fail2ban.server         [1562]: INFO    Jail 'nginx-404' reloaded
2026-11-02 19:46:25,617 fail2ban.server         [1562]: INFO    Reload finished.
2026-11-02 19:52:41,217 fail2ban.transmitter    [1562]: ERROR   Command ['get', 'sshd', 'maxretry,', 'y', '
lo', 'mismo', 'con', 'findtime', 'y', 'bantime'] has failed. Received Exception('Invalid command (no get ac
tion or not yet implemented)')
```

```
2026-11-02 20:44:03,887 fail2ban.actions        [1562]: INFO      banTime: 3600
2026-11-02 20:44:03,931 fail2ban.filter         [1562]: INFO      encoding: UTF-8
2026-11-02 20:44:04,012 fail2ban.server         [1562]: INFO    Reload jail 'nginx-404'
2026-11-02 20:44:04,013 fail2ban.filter         [1562]: INFO      maxRetry: 5
2026-11-02 20:44:04,013 fail2ban.filter         [1562]: INFO      findtime: 600
2026-11-02 20:44:04,013 fail2ban.actions        [1562]: INFO      banTime: 3600
2026-11-02 20:44:04,014 fail2ban.filter         [1562]: INFO      encoding: UTF-8
2026-11-02 20:44:04,020 fail2ban.server         [1562]: INFO    Jail 'sshd' reloaded
2026-11-02 20:44:04,035 fail2ban.server         [1562]: INFO    Jail 'nginx-404' reloaded
2026-11-02 20:44:04,038 fail2ban.server         [1562]: INFO    Reload finished.
isabela@isabela-VirtualBox:~$ sudo ufw status verbose
Estado: activo
Acceso: on (low)
Predeterminado: deny (entrantes), allow (salientes), disabled (enrutados)
Perfiles nuevos: skip

Hasta                      Acción      Desde
-----                      ------      -----
2224/tcp                   ALLOW IN    Anywhere
5405/udp                   ALLOW IN    Anywhere
80/tcp                     ALLOW IN    Anywhere
2049/tcp                   ALLOW IN    Anywhere
22/tcp (OpenSSH)           ALLOW IN    Anywhere
22/tcp                     ALLOW IN    Anywhere
2224/tcp (v6)              ALLOW IN    Anywhere (v6)
5405/udp (v6)              ALLOW IN    Anywhere (v6)
80/tcp (v6)                ALLOW IN    Anywhere (v6)
2049/tcp (v6)              ALLOW IN    Anywhere (v6)
22/tcp (OpenSSH (v6))      ALLOW IN    Anywhere (v6)
22/tcp (v6)                ALLOW IN    Anywhere (v6)
```

```
22/tcp (OpenSSH (v6))      ALLOW IN    Anywhere (v6)
22/tcp (v6)                ALLOW IN    Anywhere (v6)

isabela@isabela-VirtualBox:~$ systemctl is-active ssh nginx fail2ban
active
active
active
isabela@isabela-VirtualBox:~$ sudo sshd -T | grep -Ei "permitrootlogin|maxauthtries|logingracetime"
logingracetime 30
maxauthtries 3
permitrootlogin no
isabela@isabela-VirtualBox:~$
```